

FairLens AI Bias Audit Report

Audit ID: FL-2026-259588

Date: May 03, 2026

Submitted by: CSV Upload User

Executive Summary

This audit assessed an AI-powered loan approval system for bias, discrimination, and regulatory compliance. The system received an overall risk rating of **HIGH RISK** ■ with a composite fairness score of **65.0/100**.

The model shows moderate bias concerns that should be addressed before deployment. Remediation actions are recommended within 90 days.

Regulatory Context

This audit evaluates compliance with:

- **Equal Credit Opportunity Act (ECOA):** Prohibits discrimination in credit decisions
- **EU AI Act Article 9:** Classifies credit scoring as high-risk AI requiring strict oversight
- **Fair Credit Reporting Act (FCRA):** Regulates use of consumer credit information
- **GDPR Article 5:** Requires data minimization and privacy protection
- **Federal Reserve SR 11-7:** Model risk management guidance

1. Discrimination Detection Results

■ *Disparate Impact Analysis*

Disparate Impact Ratio: 0.65

Statistical Parity Difference: 0

Equal Opportunity Difference: 0

Interpretation:

Analyzed 3 protected attributes. Minimum DI ratio: 0.650

■ *Gender DI Analysis*

Disparate Impact Ratio: 0.97

Group Approval Rates:

- Female: 52.51% (reference/highest)
- Male: 50.7% (lowest)

DI = 0.97 ■

■ *Ethnicity DI Analysis*

Disparate Impact Ratio: 0.93

Group Approval Rates:

- White: 54.03% (reference/highest)
- Hispanic: 51.37%
- Asian: 51.09%
- Black: 51.01%
- Other: 50.45% (lowest)

DI = 0.93 ■

■ *Zip_Code DI Analysis*

Disparate Impact Ratio: 0.65

Group Approval Rates:

- 10002: 61.06% (reference/highest)
- 10000: 60.64%
- 10003: 57.69%
- 10009: 53.19%
- 10006: 52.1%
- 10005: 51.52%
- 10001: 48.45%

- 10007: 44.32%
- 10004: 44.23%
- 10008: 39.77% (lowest)

DI = 0.65 ■

2. Accountability & Governance

Governance Maturity Level: Defined (Level 2)

Maturity Score: 60/100

Risk Level: ■ MEDIUM

RACI Matrix

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Model Development	Data Science Team	CTO	Compliance Dept	CEO
Bias Testing & Validation	Compliance Officer	CRO	Legal Dept	Board of Directors
Regulatory Approval	CRO	CEO	Compliance Dept	Regulators
Production Deployment	Engineering Team	CTO	Risk Management	CEO
Ongoing Monitoring	Risk Control Team	CRO	Compliance Dept	CEO
Complaint Investigation	Compliance Officer	CRO	Legal Dept	Customer Service
Annual Re-audit	External Auditors	Board of Directors	Compliance Dept	Regulators

3. Data Privacy Compliance

GDPR Compliance Checklist

- Data source documented and authorized
- Data minimization principle applied
- Retention period defined
- Regular privacy impact assessment recommended
- Data subject rights procedure should be documented

4. Remediation Recommendations

■ **HIGH PRIORITY - Action Required within 90 days**

- **Finding:** Primary Protected Attribute shows discrimination (DI=0.65)
- **Action:** Remove or replace feature in next model update
- Retrain model without this feature
- Re-run bias testing after retraining
- **Timeline:** 90 days

■ **COMPLIANT - Continue Monitoring**

- **Finding:** Gender within acceptable range (DI=0.97)
- **Action:** Maintain quarterly review schedule
- Document compliance status
- **Timeline:** Ongoing

■ **COMPLIANT - Continue Monitoring**

- **Finding:** Ethnicity within acceptable range (DI=0.93)
- **Action:** Maintain quarterly review schedule
- Document compliance status
- **Timeline:** Ongoing

■ **HIGH PRIORITY - Action Required within 90 days**

- **Finding:** Zip_Code shows discrimination (DI=0.65)
- **Action:** Remove or replace feature in next model update
- Retrain model without this feature
- Re-run bias testing after retraining
- **Timeline:** 90 days

Audit Trail

Audit ID: FL-2026-259588

Audit Date: 2026-05-03 12:06:25 UTC

Submitted By: CSV Upload User

Model Version: 1.0.0

Next Review Date: 2026-08-03

Composite Risk Score: 65.0/100

Overall Risk Level: HIGH RISK ■

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